

Class 12: Introduction to Genome Informatics

Erin Jang (PID: A17713992)

Table of contents

Section 1. Proportion of G/G in a population	1
Section 4: Population Scale Analysis (Homework)	2

Section 1. Proportion of G/G in a population

Downloaded a CSV file from Ensembl https://www.ensembl.org/Homo_sapiens/Variation/Sample?db=core;r=17:39895045-39895145;v=rs8067378;vdb=variation;vf=959672880;sample=MXL#373531_tablePanel

Here we read this CSV file:

```
mxl <- read.csv("373531-SampleGenotypes-Homo_sapiens_Variation_Sample_rs8067378.csv")
head(mxl)
```

	Sample...	Male.Female.Unknown.	Genotype..forward.strand.	Population.s.	Father
1		NA19648 (F)		A A ALL, AMR, MXL	-
2		NA19649 (M)		G G ALL, AMR, MXL	-
3		NA19651 (F)		A A ALL, AMR, MXL	-
4		NA19652 (M)		G G ALL, AMR, MXL	-
5		NA19654 (F)		G G ALL, AMR, MXL	-
6		NA19655 (M)		A G ALL, AMR, MXL	-
	Mother				
1	-				
2	-				
3	-				
4	-				
5	-				
6	-				

```
table(mx1$Genotype..forward.strand.)
```

```
A|A A|G G|A G|G
 22 21 12  9
```

```
table(mx1$Genotype..forward.strand.) / nrow(mx1) * 100
```

```
      A|A      A|G      G|A      G|G
34.3750 32.8125 18.7500 14.0625
```

Section 4: Population Scale Analysis (Homework)

One sample is obviously not enough to know what is happening in a population. You are interested in assessing genetic differences on a population scale. So, you processed about ~230 samples and did the normalization on a genome level. Now, you want to find whether there is any association of the 4 asthma-associated SNPs (rs8067378...) on ORMDL3 expression.

How many samples do we have?

```
expr <- read.table("rs8067378_ENSG00000172057.6.txt")
head(expr)
```

```
      sample geno      exp
1 HG00367  A/G 28.96038
2 NA20768  A/G 20.24449
3 HG00361  A/A 31.32628
4 HG00135  A/A 34.11169
5 NA18870  G/G 18.25141
6 NA11993  A/A 32.89721
```

```
nrow(expr)
```

```
[1] 462
```

Q13: Read this file into R and determine the sample size for each genotype and their corresponding median expression levels for each of these genotypes.

```
table(expr$geno)
```

```
A/A A/G G/G  
108 233 121
```

```
median(expr$exp[expr$geno == "A/A"])
```

```
[1] 31.24847
```

```
median(expr$exp[expr$geno == "A/G"])
```

```
[1] 25.06486
```

```
median(expr$exp[expr$geno == "G/G"])
```

```
[1] 20.07363
```

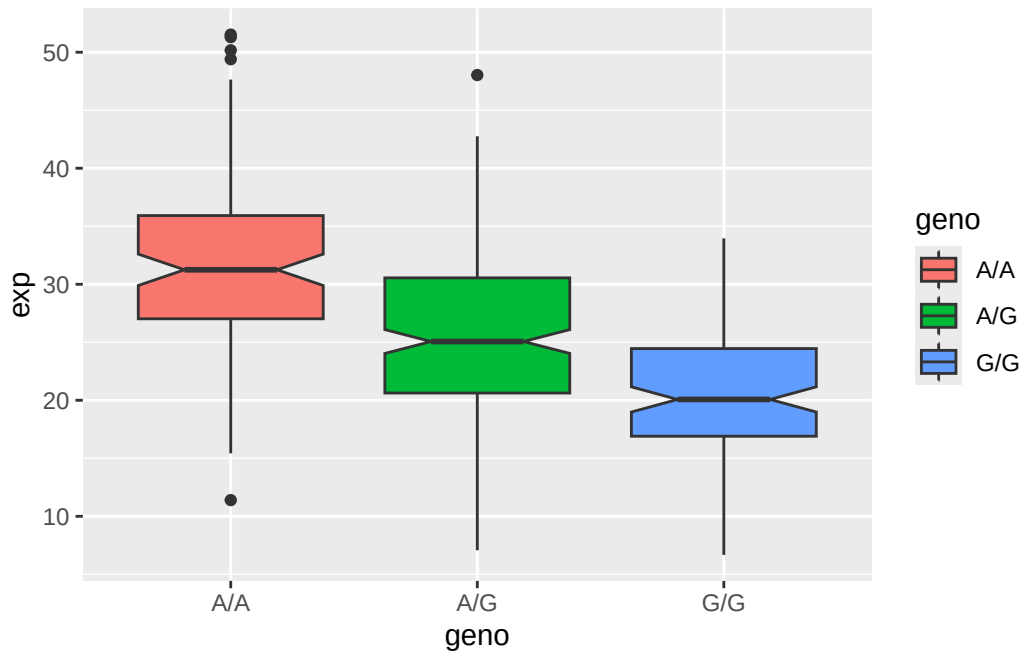
The sample size for A/A genotype is 108, 233 for A/G genotype, and 121 for G/G genotype. Their median expression levels for the A/A, A/G, and G/G genotypes are 31.25, 25.06, and 20.07, respectively.

Q14: Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORMDL3?

Let's make a boxplot:

```
library(ggplot2)
```

```
boxplot <- ggplot(expr) + aes(geno, exp, fill = geno) +  
  geom_boxplot(notch = TRUE)  
boxplot
```



The boxplot shows that individuals with the A/A genotype have much higher ORMDL3 expression levels compared to individuals with the G/G genotype. This suggests that the SNP is associated with differences in ORMDL3 expression, supporting that the SNP affects ORMDL3 gene expression.